Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Work out the value of $\dfrac{3^7 \times 3^{-2}}{3^3}$

(Total for Question 1 is 2 marks)

2 $v^2 = u^2 + 2as$

$u = 12 \quad a = -3 \quad s = 18$

(a) Work out a value of v.

$v^2 = 12^2 + 2 \times (-3) \times 18$

$v^2 = 144 - 108 = 36$

$v = 6$ is one value. (-6 also works.)

1 mark
Substitute, including the negative sign

1 mark
One valid square root is enough

6

(2)

(b) Make s the subject of $v^2 = u^2 + 2as$

$v^2 = u^2 + 2as$

$-u^2 \qquad -u^2$

$v^2 - u^2 = 2as$

$\div 2a \qquad \div 2a$

$s = \dfrac{v^2 - u^2}{2a}$

1 mark
Subtract u^2 from both sides

1 mark
Divide by 2a; a must be nonzero

(2)

(Total for Question 2 is 4 marks)

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